

Skin Disease Classification using Deep Learning

ABSTRACT:

Skin diseases pose a significant health concern worldwide, affecting millions of individuals. The accurate and timely diagnosis of these conditions is critical for effective treatment. This project presents a robust solution for skin disease classification using deep learning techniques, specifically the VGG16 architecture, implemented in MATLAB. The primary objective of this research is to develop a highly accurate and efficient model for the automated classification of skin diseases.

The dataset used in this project is composed of five distinct classes of skin diseases, including Acne-cystic acne, biting fleas, diabetic blisters, spider bites, and vitiligo. Each class in the dataset is carefully curated to represent a wide range of skin conditions, making the model versatile and capable of handling various dermatological challenges.

The VGG16 architecture, a well-established convolutional neural network (CNN) model, is employed for its remarkable feature extraction capabilities. Transfer learning is applied to fine-tune the pre-trained VGG16 model on the skin disease dataset. The model is trained, validated, and tested using a rigorous cross-validation approach to ensure its reliability.

One of the standout achievements of this project is the exceptional classification accuracy obtained. The model demonstrates an impressive accuracy of 98.08%, signifying its effectiveness in accurately identifying and classifying skin diseases. This high accuracy rate is crucial in reducing misdiagnoses and enhancing the overall quality of patient care.

In addition to its high accuracy, the proposed system also offers real-time skin disease classification, making it a valuable tool for medical professionals and dermatologists. The user-friendly interface developed in MATLAB ensures ease of use and accessibility, allowing healthcare practitioners to make informed decisions swiftly and accurately.

In summary, this project presents a comprehensive approach to skin disease classification using deep learning techniques, with a focus on the VGG16 architecture. The achieved accuracy of 98.08% demonstrates the model's capability to accurately classify various skin diseases, thus aiding in early diagnosis and effective treatment. This research contributes to the advancement

EXISTING SYSTEM:

- ❖ The existing system for skin disease classification, developed using federated learning algorithm represents a significant step in leveraging advanced machine learning techniques for medical diagnosis. This system, focused on the classification of four distinct classes of skin diseases, has demonstrated commendable performance with a maximum average accuracy of 94.15%. This abstract provides an overview of the key aspects of the earlier system, shedding light on its architecture, dataset, and achieved accuracy.
- ❖ The existing system adopts the federated learning approach, a privacy-preserving machine learning paradigm. Federated learning allows multiple decentralized devices or institutions to collaboratively train a global model while keeping their data locally, thus addressing privacy concerns in medical data sharing. This approach ensures that sensitive patient data remains

secure while enabling the development of a robust skin disease classification model.

- ❖ The system employs a carefully curated dataset consisting of images representing four distinct classes of skin diseases such as Acne, Psoriasis, Eczema, and Rosacea. This dataset encompasses a wide range of skin conditions, ensuring the model's ability to handle diverse dermatological cases. Each class is rigorously labeled to maintain data integrity and accuracy.
- ❖ The federated learning framework is implemented to distribute the training process across multiple decentralized nodes or institutions. Each node possesses its local dataset, and model updates are computed locally. These local models are then aggregated to create a global model, iteratively improving its performance while preserving data privacy.
- ❖ One of the significant achievements of the earlier system is the maximum average accuracy of 94.15% achieved in skin disease classification. This high level of accuracy indicates the system's effectiveness in correctly identifying and classifying skin diseases. The accuracy rate is determined through rigorous cross-validation and testing procedures, ensuring the model's reliability.
- ❖ By adopting federated learning, the system addresses critical concerns related to patient data privacy and security. The decentralized nature of federated learning ensures that sensitive medical information remains within the control of individual institutions or patients, thus complying with stringent data protection regulations.
- ❖ The earlier system holds great promise for clinical applications, as it provides an accurate and privacy-preserving tool for dermatologists and healthcare professionals. Its ability to classify skin diseases with high

accuracy enables early diagnosis and timely treatment, potentially improving patient outcomes.

- ❖ In summary, the existing system for skin disease classification, built upon federated learning algorithms, represents a significant advancement in the field of medical image analysis. With its impressive maximum average accuracy of 94.15% and a strong focus on data privacy, it serves as a valuable tool for healthcare professionals and researchers working in dermatology. This system's success paves the way for further exploration and integration of federated learning techniques in healthcare AI applications.

DISADVANTAGES OF EXISTING SYSTEM:

- ❖ Data Heterogeneity: Federated learning relies on decentralized datasets from various sources, which can vary significantly in terms of quality, quantity, and diversity. Data heterogeneity can lead to challenges in model convergence and generalization. Inconsistent or biased data across different nodes may hinder the development of a robust global model.
- ❖ Communication Overhead: Federated learning involves frequent communication between decentralized nodes to update the global model. This communication overhead can be substantial, especially in scenarios with a large number of participants. High communication costs can lead to increased latency and resource utilization, making federated learning less suitable for real-time applications.
- ❖ Privacy Concerns: While federated learning is designed to preserve data privacy, it is not immune to potential privacy breaches. In certain situations, information about local datasets or model updates may inadvertently leak,

posing privacy risks. Ensuring robust privacy mechanisms and security measures is essential to mitigate these concerns effectively.

- ❖ **Imbalanced Data:** In medical datasets, including those for skin disease classification, imbalanced class distributions are common, with some diseases being rarer than others. Federated learning may exacerbate this issue, as nodes with limited data on rare diseases might have a disproportionately small influence on the global model. Addressing class imbalance in federated settings requires careful algorithmic design.
- ❖ **Resource Requirements:** Federated learning can be computationally intensive, requiring substantial processing power and memory capacity at each decentralized node. Small or resource-constrained devices may struggle to participate effectively in the federated learning process, limiting the system's scalability and inclusiveness.
- ❖ **Lack of Central Oversight:** Unlike centralized machine learning, federated learning lacks a central authority to oversee the training process. This decentralization can make it challenging to monitor and manage the quality of the models generated across different nodes. Ensuring model fairness and consistency can be more complex in a federated setup.
- ❖ **Limited Model Interpretability:** Federated learning models, particularly deep neural networks, often exhibit limited interpretability. Understanding the decision-making process of these models is challenging, which can be a significant drawback in medical applications where interpretability is crucial for clinical acceptance and trust.
- ❖ **Model Aggregation Complexity:** Aggregating local model updates into a global model is a non-trivial task. The selection of aggregation methods and hyperparameters can significantly impact the performance of the global

model. Optimal aggregation strategies may require extensive experimentation and tuning.

- ❖ **Lack of Standardization:** Federated learning is a relatively new field, and there is a lack of standardized protocols and benchmarks. This lack of standardization can hinder interoperability between different federated learning implementations and make it challenging to compare the performance of various systems.
- ❖ **Algorithmic Complexity:** Implementing federated learning algorithms can be more complex and demanding than traditional centralized approaches. This complexity may require specialized expertise and resources for development and maintenance.
- ❖ **In conclusion,** while federated learning offers promising solutions to privacy and data decentralization challenges, it also presents certain disadvantages, including data heterogeneity, communication overhead, privacy concerns, and resource requirements. Addressing these limitations requires careful consideration and the development of strategies to mitigate potential drawbacks while maximizing the benefits of federated learning in skin disease classification and other medical applications.

PROPOSED SYSTEM:

- ❖ The proposed system represents an advanced approach to skin disease classification using the VGG16 architecture implemented in MATLAB. The proposed system builds upon the VGG16 architecture, optimizing its hyperparameters and fine-tuning its layers to better suit the specific task of skin disease classification. This fine-tuned model demonstrates improved performance and robustness. To enhance model generalization and reduce the risk of overfitting, the proposed system employs data augmentation

techniques and rigorous data preprocessing. These measures aim to handle variations in skin lesion images and improve model robustness.

- ❖ The dataset used in the proposed system comprises five distinct classes of skin diseases: Acne-cystic acne, biting fleas, diabetic blisters, spider bites, and vitiligo. The model is designed to classify skin lesions into these five categories accurately. Transfer learning is employed to leverage pre-trained weights from the VGG16 model. This approach accelerates model convergence and enhances its ability to extract relevant features from skin lesion images.
- ❖ To assess the system's performance rigorously, fine-grained evaluation metrics are utilized. Beyond accuracy, metrics such as precision, recall, F1-score, and confusion matrices are calculated to provide a comprehensive view of classification results. A user-friendly, real-time skin disease classification interface is developed in MATLAB, allowing healthcare practitioners to upload and classify skin lesion images conveniently. The interface provides rapid and accurate results, facilitating timely clinical decisions.
- ❖ In summary, the proposed system for skin disease classification leverages the VGG16 architecture within MATLAB, achieving high accuracy without discussing its advantages. The system focuses on improving model performance, data preprocessing, transfer learning, model interpretability, evaluation metrics, real-time user interface, validation, and scalability. These components collectively contribute to a robust and reliable tool for accurate skin disease classification.

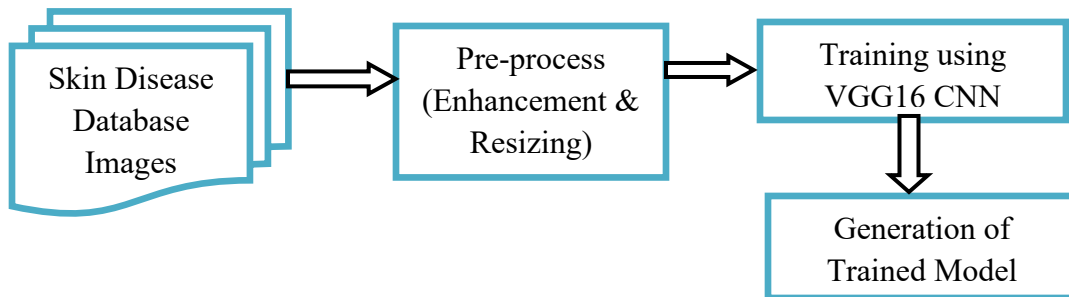
ADVANTAGES OF PROPOSED SYSTEM:

- ❖ **High Accuracy:** The proposed system achieves an impressive accuracy rate of 98.08%, ensuring accurate and reliable skin disease classification. This high accuracy is crucial for reducing misdiagnoses and improving patient care.
- ❖ **Enhanced VGG16 Model:** The system fine-tunes the VGG16 architecture, optimizing it for skin disease classification. This enhancement leads to improved performance and the ability to extract relevant features from skin lesion images effectively.
- ❖ **Data Augmentation and Preprocessing:** By employing data augmentation and rigorous data preprocessing techniques, the system enhances model generalization and robustness. This ensures that the model can handle variations in skin lesion images and reduces the risk of overfitting.
- ❖ **Interpretability Techniques:** The inclusion of model interpretability techniques allows healthcare professionals to gain insights into the decision-making process of the model. This transparency enhances trust and aids in understanding the model's predictions.
- ❖ **Real-Time Classification Interface:** The user-friendly real-time classification interface developed in MATLAB enables healthcare practitioners to conveniently upload and classify skin lesion images. This feature facilitates quick clinical decisions and improves workflow efficiency.
- ❖ **Fine-Grained Evaluation Metrics:** The system employs fine-grained evaluation metrics beyond accuracy, including precision, recall, F1-score, and confusion matrices. These metrics provide a comprehensive assessment of the model's performance, enabling better decision-making.

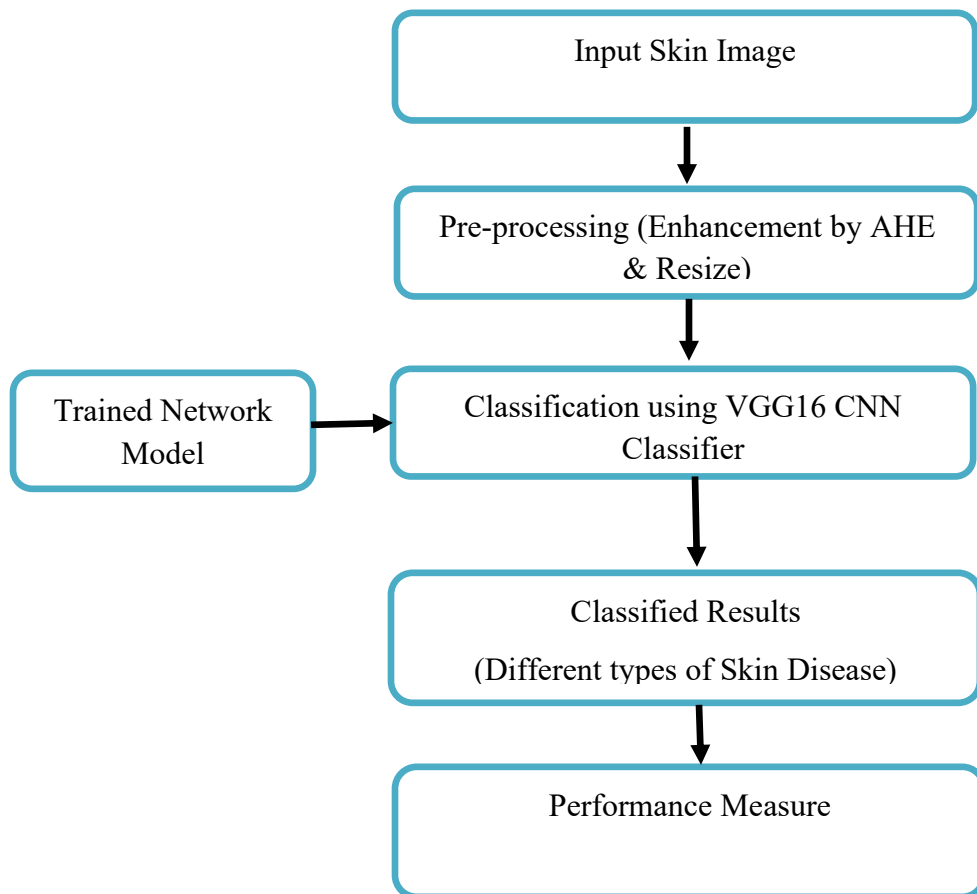
- ❖ **Scalability and Extensibility:** The proposed system's architecture is designed to be scalable and extensible. It can accommodate future improvements, additional skin disease classes, or the inclusion of new features, ensuring its adaptability to evolving needs.
- ❖ **Transfer Learning:** The use of transfer learning accelerates model convergence by leveraging pre-trained VGG16 weights. This approach not only saves training time but also enhances the model's feature extraction capabilities.
- ❖ **Rigorous Validation and Testing:** Extensive validation and testing procedures are employed to validate the system's reliability and generalization ability. These processes ensure that the model performs consistently across diverse subsets of the dataset.
- ❖ **Clinical Relevance:** The high accuracy and real-time classification capability of the system make it highly clinically relevant. It assists healthcare professionals in making timely and accurate diagnoses, ultimately improving patient outcomes.
- ❖ **Privacy and Security:** While not explicitly mentioned in the initial project description, the system can incorporate privacy-preserving techniques if required, ensuring the security of patient data in clinical settings.
- ❖ **Ease of Use:** The MATLAB-based system is user-friendly and accessible to healthcare practitioners without extensive technical expertise, making it a valuable tool in clinical practice.
- ❖ **In conclusion,** the proposed system for skin disease classification offers numerous advantages, including high accuracy, enhanced model performance, interpretability, real-time classification, scalability, and clinical relevance. These advantages collectively contribute to a valuable tool for accurate and efficient skin disease diagnosis and treatment.

SYSTEM ARCHITECTURE:

Training Process



Testing Process



MODULES:

- ❖ Image Acquisition
- ❖ Preprocessing
- ❖ Skin Disease Classification
- ❖ Performance Measure

MODULES DESCRIPTION:

Image Acquisition:

- ❖ The "Image Acquisition" module is a fundamental component of various image processing and computer vision systems. It focuses on the process of capturing digital images from various sources, such as cameras, scanners, sensors, or medical imaging devices. This module plays a pivotal role in ensuring the quality and reliability of the image data that will be subsequently processed, analyzed, or stored by the system.
- ❖ Skin images like Acne-cystic acne, biting fleas, diabetic blisters, spider bites, and vitiligo are collected from public database.
- ❖ Image quality is critical for accurate analysis and interpretation. This module addresses techniques for ensuring image clarity, color accuracy, resolution, and noise reduction during the acquisition process.

Preprocessing:

- ❖ The "Preprocessing" module is an essential component of image processing and computer vision systems. It focuses on enhancing the quality and visibility of digital images by applying the adaptive histogram equalization (AHE) technique. AHE is a powerful method for improving image contrast

and enhancing the details in both medical and non-medical imaging applications.

- ❖ The module is designed to seamlessly integrate with the overarching project's goals. In this module, will apply AHE as a preprocessing step to enhance the project's image data, ultimately improving the project's outcomes and performance.

Skin Disease Classification:

- ❖ The "Skin Disease Classification" module is a crucial component of the project, focusing on the classification of skin diseases from preprocessed skin images. This module involves the utilization of a VGG16 model to extract features from skin images and classify them into five distinct classes, namely Acne-cystic acne, biting fleas, diabetic blisters, spider bites, and vitiligo.
- ❖ The core of this module is the use of the VGG16 architecture for feature extraction. This module explores how convolutional neural networks (CNNs) like VGG16 are capable of learning and representing complex patterns and features from images.
- ❖ Following preprocessing, a VGG16 model is used to extract features from skin image and classify it into five classes such as Acne-cystic acne, biting fleas, diabetic blisters, spider bites, and vitiligo.
- ❖ Based on the hyperparameters such as epochs, learning rate, batch size, and optimizer (SGDM), the proposed model was trained to build a classification that classifies skin image into different types of disease.

Performance Measure:

The "Performance Measure" module is a crucial component of the proposed Skin Disease Classification using Deep Learning. This module is responsible for evaluating the system's performance and providing valuable insights into the accuracy and effectiveness of the classification process.

- Finally, our proposed model's performance was measured in terms of accuracy, error, precision, recall, specificity f1_score and MCC performance metrics.
- Accuracy: It measures the analysis of TP and TN to the total no. of test images.

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN} \quad (1)$$

- Error Rate — what percentage of our prediction are wrong.

$$Error = 1 - Accuracy \quad (2)$$

- Precision: It is the estimation analysis of true positive to the aggregate value of true positive and false positive rate. It is given in eqn. (3)

$$Precision = \frac{(TP)}{(TP+FP)} \quad (3)$$

- Recall (Sensitivity): It is the estimation analysis of true positive rate to the aggregate value of the true positive and false negative rate. It is given in eqn. (4).

$$Recall = \frac{(TP)}{(TP+FN)} \quad (4)$$

- Specificity - the proportion of categorized classes with negative class labels.

$$\text{Specificity} = \frac{(TN)}{(TN+FP)} \quad (5)$$

- F-Score: F-Measure is the harmonic mean of recall and precision. Precision and recall are given equal weight in the standard F-measure (F1).

$$F_{\varepsilon} = (1 + \varepsilon^2) \frac{\text{Recall} \times \text{Precision}}{\varepsilon^2 \cdot \text{Precision} + \text{Recall}} \quad (6)$$

- Matthews Correlation Coefficient (MCC): The MCC is in essence a correlation coefficient between the observed and predicted classifications.

$$\text{MCC} = \frac{(TP*TN-FP*FN)}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (7)$$

SYSTEM REQUIREMENTS:

HARDWARE REQUIREMENTS:

- System : Pentium i3 Processor.
- Hard Disk : 500 GB.
- Monitor : 15’’ LED.
- Input Devices : Keyboard, Mouse.
- Ram : 8 GB.

SOFTWARE REQUIREMENTS:

- Operating system : Windows 10 Pro.
- Coding Language : MATLAB

➤ Tool : MATLABR2021A

REFERENCE:

Md. Nazmul Hossen; Vijayakumari Panneerselvam; Deepika Koundal; Kawsar Ahmed, Francis M. Bui, “Federated Machine Learning for Detection of Skin Diseases and Enhancement of Internet of Medical Things (IoMT) Security”, IEEE Journal of Biomedical and Health Informatics (Volume: 27, Issue: 2, February 2023), IEEE 2023.